Thrombocytopenia 6 (THC6) — Comprehensive Disease Characterization

Causal gene: *SRC* (proto-oncogene, non-receptor tyrosine-protein kinase c-Src; HGNC:11283; NCBI Gene 6714; UniProt P12931). Thrombocytopenia 6 is a Mendelian, autosomal-dominant inherited platelet disorder caused by a recurrent **germline gain-of-function missense variant in *SRC*, c.1579G>A p.(Glu527Lys) (E527K)**. It is distinct from the many other numbered "thrombocytopenia" entries (e.g., THC2/ANKRD26, THC5/ETV6), which involve different genes; this report follows the identifiers **OMIM:616937 / MONDO:0014837 / causal gene SRC**.

> Evidence base: The disease is defined by a small number of primary human reports — Turro et al. 2016 (P 26936507), landmark discovery, 9 cases), De Kock et al. 2019 (P 31204551), second family, de novo), and Palma-Barqueros et al. 2022 (P 35349645), third/additional kindred) — supplemented by SRC platelet/megakaryocyte reviews (De Kock & Freson 2020, P 33255186) and SRC mouse bone-biology studies (P 7689855, P 8634424). Evidence types are flagged throughout as *human clinical*, *in vitro*, *model organism*, or *computational*.

1. Disease Information

Overview. THC6 (SRC-related thrombocytopenia, SRC-RT) is an ultra-rare, autosomal-dominant, syndromic **inherited macro-thrombocytopenia**. The phenotype ranges from isolated mild thrombocytopenia to a complex multisystem disease combining bleeding, dysmorphic α-granule–poor giant platelets, **early-onset myelofibrosis**, splenomegaly, bone pathology, and (in some carriers) immunodeficiency and neurological features (P 26936507 ; P 35349645). It is notable as the **first reported deleterious germline mutation of SRC**, a classic proto-oncogene (*human clinical / in vitro*; P 26936507).

Key identifiers. - OMIM: **616937** (THROMBOCYTOPENIA 6; THC6) - MONDO: **MONDO:0014837** - Gene: **SRC**, HGNC:11283, NCBI Gene 6714, Ensembl ENSG00000197122, cytoband **20q11.23** - Orphanet: ultra-rare inherited thrombocytopenia (no stable prevalence code; part of the "constitutional thrombocytopenia" group) - ICD-11: **3B64.1Z** / **3B64** (constitutional/hereditary thrombocytopenia, unspecified); ICD-10: **D69.42** (congenital and hereditary thrombocytopenia) — mapping is approximate (no THC6-specific code) - MeSH: closest headings "Thrombocytopenia" (D013921), "Blood Platelet Disorders," "Primary Myelofibrosis"

Synonyms. SRC-related thrombocytopenia (SRC-RT); THC6; hereditary thrombocytopenia with early-onset myelofibrosis.

Information source. Disease-level knowledge is derived from **individual-patient case series** (not aggregated EHR registries), because the disorder is ultra-rare with <20 genetically confirmed patients reported worldwide.

2. Etiology

Primary cause — genetic. THC6 is a monogenic disorder caused by the heterozygous germline **SRC c.1579G>A, p.(Glu527Lys)** gain-of-function variant (*human clinical / in vitro*; P 26936507).

No environmental or infectious cause initiates the disease.

Genetic risk factors. - *Causal variant:* SRC p.E527K — the only recurrent pathogenic allele reported to date; classified **pathogenic** (ACMG/AMP) given de novo occurrence, functional gain-of-function data, absence from population databases, and segregation (P 26936507; P 31204551; P 35349645).

- *Susceptibility loci / modifier genes:* none established. Variable expressivity implies unidentified genetic/epigenetic modifiers. Downstream signaling partners **CSK** (C-terminal Src kinase; sets the pTyr530 brake), **BTK**, **PLCG2**, and integrin β3 (**ITGB3**) are mechanistic partners and plausible (unproven) modifier candidates.

Environmental / lifestyle risk factors. None causal. As general modifiers of bleeding risk: antiplatelet drugs (aspirin, NSAIDs), anticoagulants, trauma, and surgery increase bleeding; these are *exacerbating* rather than *causal* factors.

Protective factors. No genetic or environmental protective factors are established. Mechanistically, **SRC kinase inhibition** reverses the cellular defect in vitro (P 26936507) — a therapeutic rather than naturally protective lever.

Gene–environment interactions. Not characterized. The superimposed immune/ITP-like features in some carriers (P 35349645) could reflect gene–immune-environment interplay, but this is inferred, not demonstrated.

3. Phenotypes

Variable expressivity is a hallmark — even within one family, phenotypes range from isolated thrombocytopenia to full syndromic disease (P 35349645).

Phenotype	Type	HPO term	Onset	Severity / course	Frequency
Thrombocytopenia (platelets ~38–139 $\times 10^9$/L)	Lab abnormality	HP:0001873	Congenital/ childhood	Mild–moderate, chronic, fluctuating	Essentially all patients
Large/giant platelets, anisocytosis	Lab/cell sign	HP:0011869; ↑MPV HP:0011877	Congenital	Variable	Most
Platelet α-granule paucity	Lab (ultrastructural)	HP:0030401	Congenital	—	Reported
Mucocutaneous bleeding (bruising, epistaxis, menorrhagia, GI)	Symptom/sign	HP:0000978; HP:0000421; HP:0000132	Childhood–adult	Mild–moderate, episodic	Common
Early-onset myelofibrosis + trilineage dysplasia	Clinical/path sign	HP:0011974; HP:0012145	Childhood–adult	Progressive in subset	Subset
Splenomegaly	Clinical sign	HP:0001744	Variable	—	Subset
Bone disease (osteopenia/ osteoporosis, bone pain, fractures)	Clinical sign	HP:0000939; HP:0003025	Variable	—	Subset
Immune defects / recurrent infections	Clinical sign	HP:0002719	Variable	—	4/7 carriers in one kindred (P 35349645)
Neurological (specific language impairment, epilepsy)	Behavioral/ neuro	HP:0000750; HP:0001250	Childhood	Variable	Some family members (P 35349645)

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data exist for THC6. Expected impact: bleeding-related activity restriction, fatigue/transfusion burden if myelofibrosis progresses, and neurodevelopmental/educational impact where language impairment or epilepsy occur. Qualitative only.

---

4. Genetic / Molecular Information

- **Causal gene:** SRC (HGNC:11283; OMIM *190090; chr20q11.23). Encodes the 536-aa non-receptor tyrosine kinase c-Src with SH3, SH2, and kinase (SH1) domains plus an N-terminal myristoylation/membrane-anchor and a C-terminal regulatory tail.
- **Pathogenic variant: c.1579G>A, p.(Glu527Lys)**, NM_005417. Type: **missense**. Classification: **Pathogenic** (ACMG/AMP: PS3 functional, PS2/PM6 de novo, PM1 critical regulatory region adjacent to pTyr530, PM2 absent from gnomAD, PP1 segregation). It is the sole recurrent disease allele reported (P 26936507; P 31204551; P 35349645).
- **Allele frequency:** essentially **absent from gnomAD / 1000 Genomes / ExAC** (consistent with a highly penetrant disease allele).
- **Origin: germline**; arises both **inherited** (multigenerational kindreds) and **de novo** (P 31204551). SRC is somatically activated in many cancers, but THC6 is defined by the germline allele.
- **Functional consequence: gain of function / constitutive activation.** E527K lies immediately adjacent to the autoinhibitory C-terminal phosphotyrosine (pTyr530) and **"predicts loss of SRC's self-inhibitory capacity,"** confirmed in vitro by increased kinase activity and enhanced activation-loop Tyr419 autophosphorylation (P 26936507, *in vitro / computational*). (Note: residue numbering — E527/Y530 in the 536-aa isoform correspond to the classic E524/Y527 and avian Y416/Y527 conventions.)
- **Modifier genes:** none proven (see §2).

- **Epigenetic information:** no disease-specific methylation/chromatin data available.
- **Chromosomal abnormalities:** none; THC6 is a point-mutation disorder (no CNV/translocation association). Bone marrow may show trilineage **dysplasia** but not a defining karyotype.

---

5. Environmental Information

- **Environmental factors / toxins / radiation:** none implicated as causal.
- **Lifestyle factors:** no causal role; antiplatelet agents and alcohol can worsen bleeding tendency (general principle).
- **Infectious agents:** not a cause. Recurrent infections observed in some carriers are a *consequence* of the associated immune defect, not a trigger (P 35349645).

Not applicable / not available for most environmental sub-items — THC6 is a purely genetic disorder.

---

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Germline SRC c.1579G>A (p.E527K) mutation** occurs adjacent to the C-terminal regulatory tyrosine (pTyr530) → **leads to** loss of the intramolecular pTyr530–SH2 autoinhibitory clamp (*computational/in vitro*; P 26936507).
2. Loss of the clamp → **results in** SRC being locked in the open, catalytically active conformation with enhanced autophosphorylation at activation-loop Tyr419 → **constitutive kinase activity** (*in vitro*; P 26936507).

3. Constitutive SRC activity → **leads to** globally enhanced tyrosine phosphorylation in platelets and megakaryocytes (MKs); the active form predominates in patient platelets (*human clinical*;

   P 26936507

).
4. **Branch A (platelet-production defect, primary):** hyperactive SRC → **drives** formation of Tyr419-SRC–positive **podosomes** that **induce altered actin cytoskeletal organization** in MKs → **results in** reduced **proplatelet formation** → **thrombocytopenia** with dysmorphic, size-variable, **α-granule–poor** platelets. This step is **pharmacologically reversible by SRC inhibition** (*human clinical / in vitro*;

   P 26936507

).
5. **Branch B (marrow/fibrosis):** abnormal MK signaling in a hypercellular marrow with trilineage dysplasia → **contributes to early-onset myelofibrosis** (reticulin fibrosis) and splenomegaly / extramedullary hematopoiesis (*human clinical*;

   P 26936507

).
   Patient stem cells form more myeloid and MK colonies in vitro.
6. **Branch C (bone):** SRC is essential for **osteoclast** ruffled-border formation and bone resorption; dysregulated SRC activity → **leads to** abnormal bone remodeling and bone disease. Inferred from the central role of SRC in osteoclasts (*model organism*; Src-knockout mouse osteopetrosis,

   P 7689855

).
7. **Branch D (immune/neuro):** constitutive activation of downstream **BTK** and **PLCG2**, with increased ERK phosphorylation in B cells on sIgM crosslinking → **contributes to** immune dysregulation / recurrent infections; neurological features (language impairment, epilepsy) reflect SRC's broad CNS expression (mechanism inferred) (*human clinical*;

   P 35349645

).
8. Platelet number reduction + mild platelet-function impairment → **results in** the clinical **bleeding** phenotype (*human clinical*).

Detail by category

- **Molecular pathways:** SRC-family-kinase (SFK) signaling; ITAM/GPVI and integrin αIIbβ3 **outside-in** signaling; SRC→SYK→BTK→PLCγ2 and SRC→RAS/MAPK(ERK) cascades

(; P 33255186). P 35349645

KEGG/Reactome anchors: Reactome "GPVI-mediated activation cascade," "Integrin αIIbβ3 signaling."

- **Cellular processes:** actin cytoskeleton organization, podosome assembly, megakaryocyte maturation/proplatelet formation, bone resorption, B-cell receptor signaling.
- **Protein dysfunction:** gain-of-function via disrupted autoinhibition (not misfolding/aggregation). PDB references for SRC autoinhibited conformation: **1FMK, 2SRC**; AlphaFold AF-P12931.
- **Immune involvement:** constitutive BTK/PLCγ2/ERK activation; recurrent infections in a subset — a mixed immunodeficiency/autoimmune picture (P 35349645).
- **Tissue-damage mechanism:** marrow **fibrosis** (reticulin deposition) is the principal tissue-injury readout.
- **GO terms (BP):** GO:0018108 peptidyl-tyrosine phosphorylation; GO:0004715 non-membrane-spanning protein tyrosine kinase activity; GO:0030036 actin cytoskeleton organization; GO:0030220 platelet formation; GO:0002576 platelet degranulation; GO:0045453 bone resorption.
- **GO terms (CC):** GO:0031091 platelet alpha granule; GO:0002102 podosome; GO:0005856 cytoskeleton.
- **Cell types (CL):** CL:0000556 megakaryocyte; CL:0000233 platelet; CL:0000092 osteoclast; CL:0000236 B cell; CL:0000037 hematopoietic stem cell.
- **Molecular profiling:** no published transcriptomic/proteomic/metabolomic datasets specific to THC6; key experimental readouts are phospho-tyrosine Western blots and MK colony/proplatelet assays (P 26936507).

Upstream vs downstream: the mutation and constitutive kinase activity are **upstream**; podosome/actin dysregulation and impaired proplatelet formation are **midstream**; thrombocytopenia, myelofibrosis, bone disease, bleeding are **downstream** manifestations.

---

7. Anatomical Structures Affected

- **Organ level — primary: bone marrow** (UBERON:0002371) and the **blood/hematopoietic system** (UBERON:0000178); **bone/skeleton** (UBERON:0001474).
- **Secondary involvement: spleen** (UBERON:0002106; splenomegaly, extramedullary hematopoiesis); **skin/mucosa** (UBERON:0002097; bleeding/bruising); **immune system**; **central nervous system** (UBERON:0001017; language impairment, epilepsy — subset).
- **Body systems:** hematologic, skeletal, immune, and (variably) nervous systems.
- **Tissue/cell level:** connective/hematopoietic marrow stroma (fibrosis); target cells — **megakaryocytes** (CL:0000556), **platelets** (CL:0000233), **osteoclasts** (CL:0000092), **B lymphocytes** (CL:0000236).
- **Subcellular (GO CC):** platelet **α-granules** (GO:0031091; deficient), **podosomes** (GO:0002102), actin **cytoskeleton** (GO:0005856); plasma-membrane inner leaflet (SRC myristoyl anchor).
- **Localization / laterality:** systemic/bilateral (hematologic disease); no lateralization.

---

8. Temporal Development

- **Onset: congenital/pediatric** thrombocytopenia (present from early life), often detected incidentally; syndromic features (myelofibrosis, bone, splenomegaly) may be recognized in childhood or adulthood. Pattern is **chronic/insidious** (P 26936507 ; P 35349645).
- **Progression:** thrombocytopenia is generally **stable/fluctuating** and lifelong; **myelofibrosis can be progressive** in a subset (hence "early-onset myelofibrosis"). Platelet counts fluctuate (e.g., 38–139 $\times 10^9$/L) (P 35349645).
- **Disease course / duration:** chronic, lifelong.
- **Remission:** no spontaneous remission of the genetic disorder; apparent "responses" to corticosteroids/splenectomy reflect a superimposed immune component or spleen removal

rather than cure
(P 35349645).

- **Critical periods / intervention windows:** avoiding misdiagnosis as ITP (and thus unnecessary immunosuppression/splenectomy) is the key actionable window; perioperative and peripartum periods require bleeding-risk management.

---

9. Inheritance and Population

- **Epidemiology: ultra-rare**; <20 genetically confirmed patients from a handful of unrelated families reported worldwide
(P 35349645 :
"genetic diagnosis of the disease has only been reported in 7 patients from 3 unrelated families"). No reliable prevalence/incidence estimate exists.
- **Inheritance: autosomal dominant**
(P 35349645).
- **Penetrance:** appears **high** for thrombocytopenia; **variable expressivity** for syndromic features.
- **Expressivity: highly variable**, intra- and inter-familial (isolated thrombocytopenia → full syndrome)
(P 35349645).
- **Anticipation:** not a repeat-expansion disorder; no anticipation.
- **De novo / mosaicism: de novo occurrence documented**
(P 31204551);
germline mosaicism not specifically reported.
- **Founder effect / consanguinity:** none; recurrence reflects independent mutation of the same nucleotide, not a founder haplotype. Consanguinity irrelevant (dominant).
- **Carrier frequency:** not applicable (dominant, fully penetrant-for-thrombocytopenia; allele absent from gnomAD).

- **Demographics:** reported in European populations (British, Belgian, Spanish); no established ethnic predilection. **No sex bias** (autosomal). Age distribution: present from childhood, recognized across the lifespan.

10. Diagnostics

- **Laboratory:** CBC — thrombocytopenia with **increased MPV** and anisocytosis; peripheral smear/flow cytometry — **large-to-giant dysmorphic platelets**, **increased immature platelet fraction (>15%)**; **light transmission aggregometry** — mildly impaired function; **electron microscopy** — **α-granule paucity** (P 26936507; P 35349645). LOINC anchors: platelet count 777-3; MPV 32623-1.
- **Bone marrow biopsy:** hypercellular marrow, **reticulin myelofibrosis**, **trilineage dysplasia**, dysmorphic megakaryocytes (P 26936507).
- **Imaging:** ultrasound/CT for **splenomegaly**; skeletal imaging/DXA for bone disease.
- **Biomarker (research):** elevated active-SRC and global platelet phospho-tyrosine; constitutive BTK/PLCγ2 activation (P 35349645) — not yet standard clinical assays.
- **Genetic testing (definitive):** targeted **NGS inherited-platelet-disorder gene panels**, **whole-exome**, or **whole-genome sequencing** identifying **SRC c.1579G>A p.(Glu527Lys)**; single-gene Sanger confirmation and **cascade testing** of relatives (P 26936507; general IPD workflows P 31249973, P 42082146). CMA/karyotype/FISH are generally uninformative (point mutation).

- **Clinical criteria / differential diagnosis:** no formal society criteria; diagnosis rests on phenotype + molecular confirmation. **Differential:** immune thrombocytopenia (ITP), **MYH9-related disease** and other inherited macrothrombocytopenias, Bernard–Soulier syndrome, gray platelet syndrome (α-granule deficiency), and **primary myelofibrosis**. Distinguishing feature: family history + SRC genotype; the ITP overlap is the key pitfall (P 35349645).
- **Screening:** no population/newborn screening. **Cascade genetic screening** of at-risk relatives and **genetic counseling** are appropriate; prenatal/PGT is theoretically possible for known-variant families.

---

11. Outcome / Prognosis

- **Survival/mortality:** no formal survival statistics. Prognosis depends on **myelofibrosis** progression and bleeding severity; isolated-thrombocytopenia patients may have near-normal life expectancy, whereas progressive myelofibrosis carries greater morbidity/mortality risk (inferred from phenotype; P 26936507).
- **Morbidity/disability:** bleeding, transfusion dependence if marrow failure advances, bone disease/fractures, and neurodevelopmental impact (language impairment, epilepsy) in a subset.
- **QoL measures:** none validated for THC6.
- **Complications:** significant hemorrhage (surgery/trauma/peripartum), progression to marrow fibrosis/cytopenias, recurrent infections (immune subset), splenomegaly-related issues.
- **Recovery potential:** the underlying genetic defect is non-reversible; targeted SRC inhibition is investigational.
- **Prognostic factors:** degree of marrow fibrosis, platelet count/bleeding history, presence of syndromic features; **no validated prognostic biomarker**. Whether THC6 confers a myeloid-malignancy predisposition (as some germline platelet-disorder genes do) is **not established** and warrants surveillance.

---

12. Treatment

No disease-specific approved therapy exists; management is extrapolated from inherited thrombocytopenia practice (PMID:32527953; PMID:18278172).

- **Supportive hemostatic care (mainstay): antifibrinolytics** — tranexamic acid (CHEBI:48669; NCIT C47720) / aminocaproic acid for mucosal bleeding and procedures; **platelet transfusion** (NCIT C15360) for major bleeding or surgical cover; **avoid antiplatelet agents/NSAIDs**; desmopressin may be considered for procedures (general practice).
- **Caution — ITP overtreatment:** because SRC-RT mimics ITP, patients have received **corticosteroids** and **splenectomy** (NCIT C15431); in one kindred four adults underwent splenectomy with sustained platelet counts >75 ×10^9/L and transient corticosteroid responses — suggesting a real but incomplete immune component, while underscoring the risk of **inappropriate splenectomy** (PMID:35349645). **Correct genetic diagnosis prevents harmful over-treatment.**
- **Mechanism-targeted (investigational): SRC tyrosine kinase inhibitors** (e.g., **dasatinib**, NCIT C62491; saracatinib/AZD0530) **rescued proplatelet formation** in patient-derived and E527K-transduced megakaryocytes in vitro (PMID:26936507) — strong proof-of-concept, but **no clinical trial in THC6** has been reported. Paradox/caveat: SRC inhibitors themselves can cause thrombocytopenia/bleeding, so clinical translation would require careful dose-finding.
- **Advanced therapeutics:** gene therapy, RNA-based, or allele-selective approaches are conceptually attractive (dominant single recurrent allele) but not developed. Allogeneic **hematopoietic stem cell transplantation** is the only potentially curative option conceivable for severe marrow failure/myelofibrosis but is unreported for THC6 specifically.
- **Pharmacogenomics:** not applicable beyond the target itself.

- **Treatment strategy / personalized medicine:** genotype-guided — confirm SRC p.E527K, then prioritize conservative hemostatic management, bleeding-risk counseling, and myelofibrosis/bone surveillance; reserve SRC-inhibitor concepts for research settings.
- **NCIT terms:** C62491 Dasatinib; C1967 Tyrosine Kinase Inhibitor; C47720 Tranexamic Acid; C15431 Splenectomy; C15360 Transfusion; C15431/stem-cell C15431→C96746 HSCT.

13. Prevention

- **Primary prevention:** not possible (genetic). **Genetic counseling** for autosomal-dominant 50% transmission risk; options include prenatal diagnosis and **preimplantation genetic testing** for families with the known variant.
- **Secondary prevention: cascade genetic testing** of relatives for early identification; baseline/periodic monitoring of platelet counts, marrow (if indicated), and bone health.
- **Tertiary prevention:** prevent bleeding complications (avoid antiplatelet drugs, pre-procedural tranexamic acid/platelet cover), monitor/manage myelofibrosis and bone disease, manage infections in the immune subset.
- **Immunization / public-health / environmental measures:** standard care; no disease-specific vaccine or environmental intervention. Routine vaccination is prudent given the immune-defect subset.
- **Prophylaxis:** peri-operative/peripartum antifibrinolytics and platelet support as needed.

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** SRC is highly conserved. Orthologs: mouse *Src* (NCBI Gene 20779; Taxon 10090), rat *Src* (Taxon 10116), zebrafish *src* (Taxon 7955), with homologs across vertebrates (HomoloGene group for SRC).
- **Natural disease in other species: no naturally occurring animal homolog of THC6** is catalogued (no OMIA entry for a spontaneous SRC-E527K thrombocytopenia). Not applicable.
- **Veterinary relevance:** none established.

- **Comparative biology:** SRC's role in osteoclast-mediated bone resorption and in platelet/MK signaling is evolutionarily conserved; the opposing bone phenotypes of human GOF disease vs mouse LOF osteopetrosis illustrate SRC dosage sensitivity in the skeleton.
- **Transmission / zoonosis:** not applicable (non-infectious genetic disease).

---

15. Model Organisms

- **Zebrafish (Danio rerio; Taxon 7955) — disease model:** expression of **mutant src recapitulated patient phenotypes** in vivo, providing a gain-of-function model aligned with the human disease (*model organism*;). P 26936507
  Resource: ZFIN.
- **Mouse (Mus musculus; Taxon 10090) — mechanism models (loss-of-function):** germline **Src-knockout** mice develop **osteopetrosis** from osteoclast ruffled-border failure, persisting to 9–10 months without remission ( ); P 7689855
  Hck/Src double-knockout mice show severe osteopetrosis with extramedullary hematopoiesis, anemia, and leukopenia, revealing partial SFK functional overlap in osteoclasts ( ). P 8634424
  These are LOF (opposite perturbation to the human GOF disease) but establish SRC's non-redundant role in bone and marrow. Resources: MGI, IMPC/IMSR.
- **Cellular / in vitro models:** COS-7 overexpression of E527K SRC (kinase-activity/phospho-Tyr419 assays); **patient-derived and lentiviral E527K-transduced hematopoietic stem cells → megakaryocytes** reproducing impaired proplatelet formation, rescued by SRC inhibition ( ). P 26936507
  iPSC-MK models are a logical extension (not yet reported).
- **Model characteristics:** zebrafish GOF captures the disease direction; mouse LOF captures the bone pathway but not the thrombocytopenia direction. **Limitation:** no mouse knock-in E527K model has been reported, leaving a gap for in vivo GOF hematologic modeling.

- **Applications:** dissecting MK cytoskeletal/podosome biology, testing SRC-inhibitor rescue, and studying SRC in bone remodeling.

Evidence Summary (key PMIDs)

- **26936507** — Turro et al. 2016, *Sci Transl Med* — discovery; E527K gain-of-function; MK/platelet/bone phenotype; zebrafish & in vitro models; SRC-inhibitor rescue (landmark, *human clinical + in vitro + model organism*).
- **31204551** — De Kock et al. 2019 — second family, **de novo** SRC variant (*human clinical*).
- **35349645** — Palma-Barqueros et al. 2022 — additional 3-generation kindred; variable expressivity; ITP overlap/splenectomy; BTK/PLCγ2 activation; immune & neuro features (*human clinical*).
- **33255186** — De Kock & Freson 2020 — SRC (patho)biology in platelets/MKs (review).
- **7689855** / **8634424** — Src-knockout mouse osteopetrosis / osteoclast biology (*model organism*).
- **32527953** / **18278172** — Nurden & Nurden — inherited thrombocytopenia framework & management (reviews).

Limitations and Future Directions

- Ultra-rare (<20 patients): phenotype frequencies and natural history are imprecise; no QoL, survival, or prevalence statistics.
- Single recurrent allele studied; modifier genes explaining variable expressivity are unknown.
- No in vivo mammalian GOF knock-in model; no clinical trial of SRC inhibitors; malignancy-predisposition risk unquantified.
- Priorities: knock-in mouse / iPSC-MK models, careful SRC-inhibitor dose-finding studies, longitudinal natural-history and myelofibrosis-progression data, and clarification of the immune/neurological components.